Ram Charan Gubbala

AI Engineer | ML Engineer

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Profile Summary

AI Engineer specializing in LLM fine-tuning, RAG pipeline development, and production-grade machine learning systems. Delivered a 20% boost in chatbot task completion through advanced prompt engineering and hybrid search optimization using Weaviate. Backed by 2+ years of backend engineering experience on high-stakes insurance systems and recognized with a Best Research Paper Award at ICDCS 2022 for innovation in edge-based AI. Combines deep AI/ML expertise with strong software engineering fundamentals to build intelligent, scalable solutions that work in production.

Technical Skills

Languages: Python, Java, SQL
AI & Machine Learning: Large Language Models, LLM Fine-tuning, Retrieval-Augmented Generation (RAG), Prompt Engineering, Neural Networks, Deep Learning, NLP, Random Forest, Anomaly Detection, TensorFlow, PyTorch, Scikit-learn
Data Science & Analytics: Exploratory Data Analysis, Feature Engineering, Statistical Modeling, Cross-validation, Hyperparameter Tuning, Pandas, NumPy, Matplotlib, Seaborn, Power BI
Backend & Software Engineering: Spring Boot, Spring Framework, RESTful APIs, Microservices, Hibernate, SonarQube
Data Engineering & Cloud: Azure Data Factory, Azure Databricks, Azure Synapse, PySpark, ETL/ELT, Medallion Architecture, AWS, Azure DevOps, Docker, Kubernetes, CI/CD
Databases: PostgreSQL, Oracle, SQL Server, MongoDB, Weaviate
Tools & Platforms: Git, Jupyter, Streamlit, ServiceNow, Kibana, OpenAI API, PowerShell, HTML/CSS

Projects

ResuMatch – AI Resume Tailoring Tool | https://github.com/ramcharan1904/ResuMatch

- LLM-powered tool that automatically tailors resumes to job descriptions for improved ATS compatibility
- Built end-to-end application handling PDF parsing, job description scraping, and AI-driven keyword optimization while preserving original resume formatting
- Tools / languages**:** Python, Streamlit, OpenAI API, Prompt Engineering, PDF Parsing, Web Scraping

Wine Quality Prediction – End-to-End ML Pipeline | https://github.com/ramcharan1904/Wine-Tasting-End-to-End

- Production-ready machine learning pipeline demonstrating complete ML lifecycle including data ingestion, validation, model training, and deployment with experiment tracking using MLflow
- Designed a modular and scalable MLOps architecture with configuration-driven pipelines, reusable components, and CI/CD-ready deployment using Docker, AWS , and GitHub Actions for automation
- Tools / Technologies: Python, MLflow, Scikit-learn, Docker, AWS (EC2, ECR), GitHub Actions, YAML, Flask/FastAPI

End-to-End Big Data Engineering on Azure | https://github.com/ramcharan1904/olist_ETL_data

- Scalable cloud data pipeline for e-commerce analytics built on Azure following Medallion Architecture
- Architected Bronze, Silver, and Gold data layers integrating multiple sources including SQL and MongoDB, delivering analytics-ready datasets via Azure Synapse
- Tools / languages**:** Python, Streamlit, OpenAI API, Prompt Engineering, PDF Parsing, Web Scraping

Work Experience

AI Engineer Intern
Sep 2025 - Dec 2025

AriesView Remote

- Designed and developed AI-powered solutions for real estate property and user data queries, building a production-ready RAG (Retrieval-Augmented Generation) chatbot.
- Integrated Weaviate vector database with hybrid search and reranking mechanisms to improve information retrieval and response relevance.
- Fine-tuned LLMs on domain-specific real estate datasets and engineered dynamic prompt templates for context-aware interactions.
- Improved chatbot task completion and response quality by 20% through iterative optimization, collaboration, and strong problem-solving.

Senior Software Engineer
Sep 2021 - Dec 2023

Capgemini Bengaluru, KA

- Developed and maintained backend services for an insurance claims processing system, ensuring reliability across high-volume data workflows.
- Optimized SQL performance and built monthly reporting pipelines used by business stakeholders for operational decision-making.
- Enhanced microservices integration using Java, Spring Boot, Hibernate, and RESTful APIs to support scalable system architecture.
- Monitored application health and resolved incidents using Kibana, SonarQube, and ServiceNow, ensuring SLA adherence and minimizing business impact.

Machine Learning Intern
Jan 2021 - Jun 2021

Excel Rf Pvt Ltd. Hyderabad,TS

- Collected and preprocessed high-dimensional network traffic and packet data to build an end-to-end anomaly detection pipeline.
- Applied feature engineering techniques to structure noisy data for Neural Network modeling.
- Tuned model hyperparameters and evaluation thresholds using Python, TensorFlow/PyTorch, and Scikit-learn, reducing false positives by 15%.
- Identified malicious traffic patterns and presented model performance insights to the team, improving network security outcomes.

Education

Masters in Data Science | University of Alabama at Birmingham| Birmingham, Al — Dec 2025

Bachelors of Technology | ICFAI University | Hyderabad, TS — June 2021